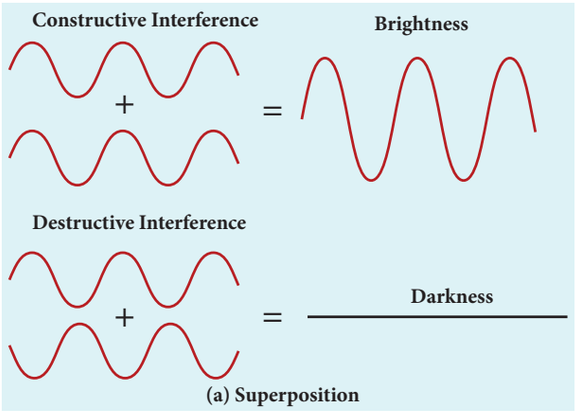


Figure 7.11 Interference due to double slit

vice-versa, the waves are out-of-phase. Hence, the displacement is minimum and these points appear dark as a result of this ***destructive interference***.

On a screen the intensity of light will be alternative maximum and minimum strips i.e. bright and dark bands which are referred as interference fringes.

7.3.4 Young's double slit experiment

Experimental setup

Thomas Young, a British Physicist in 1801 used an opaque screen with two small openings called double slit S_1 and S_2 kept equidistance from a source S as shown in Figure 7.12. The width of each slit is about 0.03 mm and they are separated by a distance of about 0.3 mm. As S_1 and S_2 are equidistant from S the same wavefront is cut by S_1 and S_2. The light waves at S_1 and S_2 are in-phase. So, S_1 and S_2 act as coherent sources which is the requirement for obtaining interference pattern.

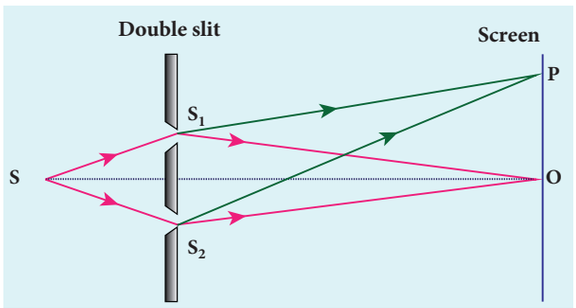


Figure 7.12 Young's double slit experimental setup

Wavefronts from S_1 and S_2 spread out and overlap on the other side of the double slit. When a screen is placed at a distance of about 1 m from the slits, alternate bright and dark fringes which are equally spaced appear on the screen. These are called interference fringes (or) bands. Using an eyepiece, the fringes can be seen directly. At the center point O on the screen, the waves from S_1 and S_2 travel equal distances and arrive in-phase as shown in Figure 7.12. These two waves constructively interfere and a bright fringe is observed at O. This is called central bright fringe. When one of the slits is closed, the fringes disappear and there is uniform illumination on the screen. This shows clearly that the bands are due to interference.

Equation for path difference

The schematic diagram of the experimental setup is shown in Figure 7.13.

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